

TAMS Tower Suitability — Numbered Analysis Reference

What you see on screen · What we fetch · What we calculate · Sub-points 1 → 12 for each scoring factor

0. When you click Analyze — overall flow

Step	What you see	What we calculate / merge
0.1	Progress bar: DEM, OSM, roads, weather, soil map	Parallel API fetch at pad / KML focus point
0.2	Map markers: pad, road snap, grid assets, shift ghosts	Haversine distances, OSRM snap, TAMS+OSM asset merge
0.3	Score card: X.X / 10 · verdict · confidence %	Weighted average of 12 factor scores
0.4	Factors panel: 12 bars with value under each	Each factor scored 0–10 from live signals
0.5	Score breakdown table: Value · Score · Weight · Contrib	Contrib = score × weight per row
0.6	GeoTech tab: Word report building / download	Separate — NOT in /10 score
0.7	Corridor panel: Suggest / Shift / Skip / Review per pad	Separate — CEA span rules, NOT in /10 score

1. Terrain slope (Weight 22%)

1.1 What you see

- Factor bar labelled **Terrain slope** with score 0–10
- Value line: e.g. 8.4° or n/a
- Note if steep: "Steep grade — pad grading / retention likely costly."
- Source tag: **Live · Open-Meteo DEM**

1.2 What we fetch

- Open-Meteo Elevation API → Copernicus DEM ~30 m
- 5 sample points: centre pad + North / South / East / West at 0.0012° (~133 m)

1.3 What we calculate

- For each offset point: run distance (m) from centre
- Rise = |elevation offset – elevation centre|
- Angle = $\text{atan}(\text{rise} / \text{run}) \times 180/\pi$
- Output slope** = maximum angle among 4 directions

1.4 Score calculation

Slope (°)	Score
≤ 5°	10 (best)
≥ 18°	0 (worst)
Between	Linear: $(18 - \text{slope}) / (18 - 5) \times 10$
Missing	5 (neutral)

1.5 Contribution to final score

Contrib = factor score × 0.22

2. Road access (Weight 14%)

2.1 What you see

- Factor bar: **Road access**
- Value: e.g. 1.24 km or live lookup failed

- Map: orange marker at nearest road snap point
- Source: **Live · OSRM nearest road**

2.2 What we fetch

- OSRM Nearest API: router.project-osrm.org/nearest/v1/driving
- Snaps pad to nearest drivable OSM road segment

2.3 What we calculate

- Road distance (km) = OSRM waypoint distance (m) ÷ 1000
- Fallback: Haversine pad → snap lat/lon if OSRM distance missing
- Stores snap coordinates for map overlay

2.4 Score calculation

Road distance	Score
≤ 0.5 km	10
≥ 8 km	0
Between	Linear interpolation
Lookup failed	3 (uncertain access)

2.5 Contribution

Contrib = factor score × 0.14

3. Water / flood buffer (Weight 14%)

3.1 What you see

- Factor bar: **Water / flood buffer**
- Value: e.g. 2.15 km, > 8 km, or fallback warning in note
- Source: **Live · OSM Overpass water** or **Fallback · Photon geocode**

3.2 What we fetch

- Primary: OSM Overpass within 8 km — natural=water, waterway, reservoir, basin
- Fallback (if Overpass fails): Photon search "lake river reservoir" — confidence –5%

3.3 What we calculate

- Nearest mapped water feature distance (km) via Haversine
- If none found within search radius → report as > 8 km
- Cap used in scoring at 8 km max

3.4 Score calculation

Water distance	Score	Meaning
≥ 0.8 km	10	Far from water = good
≤ 0.05 km (~50 m)	0	Flood / scour risk
< 0.15 km	Low	Flagged in note
Missing	5	Neutral

3.5 Contribution

Contrib = factor score × 0.14 · Also reused in factor #12

4. Settlement clearance (Weight 10%)

4.1 What you see

- Factor bar: **Settlement clearance**
- Value: e.g. 1.80 km, > 4 km
- Note if close: "Close to mapped buildings — ROW / social risk."

4.2 What we fetch

- OSM Overpass within 4 km: buildings, place nodes (city/town/village), residential landuse
- Fallback: Photon "village town" if Overpass fails

4.3 What we calculate

- Nearest building or settlement distance (km)
- Cap at 4 km for scoring curve

4.4 Score calculation

Settlement distance	Score
≥ 0.25 km	10
≤ 0.02 km	0
< 0.08 km	ROW / social risk flag
Missing	6

4.5 Contribution

Contrib = factor score $\times$ 0.10 · Also reused in factor #12

5. Power connectivity (Weight 10%)

5.1 What you see

- Factor bar: **Power connectivity**
- Value: e.g. Tower ABC · 1.2 km or none within 8 km
- Power panel: nearest tower / line / pole / substation list
- Map: grid asset markers, connection lines (straight + road route)

5.2 What we fetch (merged TAMS + OSM)

- TAMS GIS: towers, substations, transmission lines
- OSM Overpass: tower, portal, pole (≤ 15 km radius), line, cable, substation, plant
- All assets deduplicated into one **nearbyPower** list

5.3 What we calculate

- Haversine distance pad $\rightarrow$ each asset; best distance along corridor samples
- Pick nearest overall asset (tower preferred over line over SS over pole)
- Interconnect ease: ≤ 0.25 km Easy · ≤ 2 km Easy · ≤ 12 km Moderate · > 12 km Hard

5.4 Score calculation

Step	Rule
Base	Linear: 0.2 km $\rightarrow$ 10, 20 km $\rightarrow$ 0
Bonus	Pole ≤ 350 m: +2 · Tower ≤ 2 km: +1.2 · Substation: +0.8 · Easy interconnect: +0.5
Cap	Maximum score 10
No data	5 (neutral — not a rejection)

5.5 Contribution

Contrib = factor score × 0.10 · Shares merged power data with #6 and #7

6. Voltage suitability (Weight 8%)

6.1 What you see

- Factor bar: **Voltage suitability**
- Value: e.g. 220 kV, 33/110 kV, unknown
- Controls panel: selected line class kV (11–765)

6.2 What we fetch

- Voltage tags from same merged TAMS + OSM asset list (#5)
- Only tagged voltages — never invented
- Planning tiers checked: 11, 33, 66, 110, 132, 220, 400, 765 kV

6.3 What we calculate

- Available voltage set from all nearby assets
- Suggested kV: corridor asset → nearest tower → nearest line → nearest SS
- Match against useful planning tiers and high-voltage (≥33 kV) presence

6.4 Score calculation

Condition	Score
Tagged + HV ≥33 kV nearby	~9
Tagged voltages present	~7
Matched but low tier only	~6
No tags / unavailable	5 (neutral)

6.5 Contribution

Contrib = factor score × 0.08

7. Connection distance (Weight 8%)

7.1 What you see

- Factor bar: **Connection distance**
- Value: e.g. ~480 m or ~3.2 km
- Asset detail card: straight-line m + road route m (when toggled)

7.2 What we fetch

- Nearest grid asset from merged power list (#5)
- Optional OSRM driving route for map display (not used in score)

7.3 What we calculate

- Direct distance = Haversine to nearest asset (km)
- **Practical spur** = direct distance × 1.2 (screening factor for bends/ROW)

7.4 Score calculation

Practical spur	Score
≤ 0.3 km	10
≥ 15 km	0
Between	Linear interpolation
Unavailable	5

7.5 Contribution

Contrib = factor score $\times$ 0.08

8. Elevation (Weight 8%)

8.1 What you see

- Factor bar: **Elevation**
- Value: e.g. 342 m AMSL
- Same DEM fetch as factor #1

8.2 What we fetch

- Centre pad elevation from Open-Meteo 5-point grid (shared with #1)

8.3 What we calculate

- Elevation at pad centre (m above sea level)
- Band classification for logistics / flood context

8.4 Score calculation

Elevation (m)	Score
20 – 800 m	9 (favourable)
< 5 m	3 (coastal / flood plain)
> 1800 m	4 (remote logistics)
Other / missing	7 neutral

8.5 Contribution

Contrib = factor score × 0.08

9. Land cover hint (Weight 8%)

9.1 What you see

- Factor bar: **Land cover hint**
- Value: Barren / open, Vegetated, Built-up, Water, or Unknown

9.2 What we fetch

- OSM Overpass 180 m: landuse, natural, water tags
- Nominatim reverse geocode fallback if OSM tags sparse

9.3 What we calculate

- Classify into 5 categories from tag keyword matching
- Not cadastral ownership — screening hint only

9.4 Score calculation

Class	Score
Barren / open	9
Vegetated	5
Built-up	2
Water	0
Unknown	6

9.5 Contribution

Contrib = factor score × 0.08

10. Soil / SBC (Weight 8%)

10.1 What you see

- Factor bar: **Soil / SBC (open GIS + field)**
- Value: e.g. Loam · ~45% or GT-001 · 2.1 km
- GeoTech tab: full Word report, soil tables, SBC / pile calcs
- Soil card: GIS summary only (downloads in GeoTech tab)

10.2 What we fetch (one path wins)

- **Path A (preferred):** TAMS /geotech nearest field record within ~5 km
- **Path B (fallback):** ISRIC SoilGrids 2.0 — clay/sand/silt, texture, pH at depth bands

10.3 What we calculate

- Path A: score from adopted SBC (T/m^2), CBR %, design depth on file
- Path B: USDA texture class, indicative SBC range, screening confidence ~40–48%
- GeoTech report (separate): thickness-weighted layers, IS 6403 SBC, IS 2911 pile — NOT in /10

10.4 Score calculation

Source	Rule	Score
Field SBC $\geq 20 T/m^2$	Strong bearing	9
Field SBC $\geq 12 T/m^2$	Moderate	7
Field SBC lower / no SBC	On file but weak	6
SoilGrids only	$\max(5, \text{round}(\text{confidence}\% \div 10))$	5–5
Neither	Not fetched yet	5

10.5 Contribution

Contrib = factor score $\times$ 0.08

11. Wind exposure (Weight 6%)

11.1 What you see

- Factor bar: **Wind exposure**
- Value: e.g. 6.2 m/s
- Note: structural design still needs IS wind zone

11.2 What we fetch

- Open-Meteo Archive: last 90 days daily max wind @ 10 m height
- Fallback: 16-day forecast mean if archive incomplete

11.3 What we calculate

- Mean of daily max wind speeds (m/s) over available period

11.4 Score calculation

Mean wind	Score
≤ 4 m/s	10
≥ 12 m/s	0
Between	Linear interpolation
Missing	6

11.5 Contribution

Contrib = factor score $\times$ 0.06

12. Corridor feasibility (Weight 6%)

12.1 What you see

- Factor bar: **Corridor feasibility**
- Value: obstruction list e.g. close water crossing, steep terrain or No major obstruction
- Does NOT fetch new data — reuses #1, #3, #4 signals

12.2 What we reuse

- Water distance from factor #3
- Settlement distance from factor #4
- Slope degrees from factor #1

12.3 What we calculate

- Start score = 8 (optimistic baseline)
- Apply penalties for route obstacles on connection path

12.4 Score calculation

Check	Threshold	Penalty
Water proximity	< 150 m	-2
Settlement	< 100 m	-2
Steep slope	> 12°	-1.5
Very steep	> 18°	-1
Final	—	Clamped 0–10

12.5 Contribution

Contrib = factor score × 0.06

13. Final score merge (all 12 factors)

13.1 What you see

- Header: **X.X / 10** site score
- Verdict badge: Preferred · Conditional · Unsuitable
- Confidence: e.g. **72%**

13.2 What we calculate

Output	Formula
Final score	$\Sigma (\text{factor_score} \times \text{weight}) \div \Sigma \text{weights}$
Verdict	≥ 7.0 Preferred · 4.5–7.0 Conditional · < 4.5 Unsuitable
Per-row contrib	Shown in breakdown table = score $\times$ weight

14. Confidence % (partial — not all 12 factors)

14.1 What you see

- Confidence percentage on score card
- Live data panel: "X of Y signals live"

14.2 What we calculate

- Count 7 resolved signals: elevation, slope, road, water, settlement, grid (tower/SS), wind
- Base 55% + (resolved $\div$ 7) $\times$ 25% $\rightarrow$ range ~55–80%
- –5% each for Photon fallback: water, settlement, grid unavailable
- NOT counted: voltage, connection distance, land cover, soil, corridor feasibility

15. Separate from /10 score (still shown after Analyze)

#	Module	What you see	What we calculate
15.1	GeoTech Word report	Download button, soil tables, SBC/pile sheets	SoilGrids layers, IS 6403/2911 calcs — outside score
15.2	Corridor placement T1... Tn	Suggest / Shift / Skip / Review per pad	CEA span bands, min/ruling/max span, chainage
15.3	Power take-off	Best pad $\rightarrow$ nearest SS, interconnect links	Haversine + ease rating
15.4	Road route overlay	Orange driving polyline on map	OSRM route geometry — display only
15.5	Shift ghost markers	Suggested pad offset on map	Shift position along corridor spacing rules

16. Merge groups — quick lookup

Group	Factors	Shared fetch
DEM	#1 + #8	Open-Meteo 5-point elevation
Power	#5 + #6 + #7	TAMS + OSM nearbyPower merge
Soil	#10	TAMS geotech OR SoilGrids
Corridor heuristic	#12 uses #1+#3+#4	Reused distances / slope
Final	All #1–#12	Weighted average

Open-data screening only. Not utility approval, ROW certification, or foundation design. Field test values appear only when entered in TAMS /geotech.